

Knowledge graph for aligning active substance approval criteria with new chemical classifications in plant protection products.

Life sciences × Regulatory compliance and reporting × Knowledge graph and semantic modelling · OS173 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
41 is the world moving	32 can we play, can we win	LOW 0 named in the evidence	€98.6m modelled evidence · per year	NEXT regulation adopted or propose...	L1 1 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 0.0 published reference(s) in Life sciences, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity helps agrochemical companies align their active substance approval dossiers with new EU chemical classification rules. It uses a knowledge graph to map regulatory criteria to product data, reducing compliance risk. The need is urgent because the regulation is adopted but not yet applicable, giving firms a narrow window to prepare.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€119m €22.2m – €562m	€98.6m €18.4m – €467m	€394k €74k – €1.9m	modelled
Observed: tendered contract value, annualised	€19.1m	€19.1m	€77k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Life sciences (EU27_2020)	6,493 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Knowledge graph and semantic modelling	29.3 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_ain2 · E_AI_TTM	2025
Annual value of one engagement	€260k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	1,558 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.4 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Artificial intelligence by NACE Rev. 2 activity series E_AI_TTM is used as a proxy for Knowledge graph and semantic modelling: Text and language analysis as the nearest series. Contract value is a configured assumption, not observed. Only 1 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

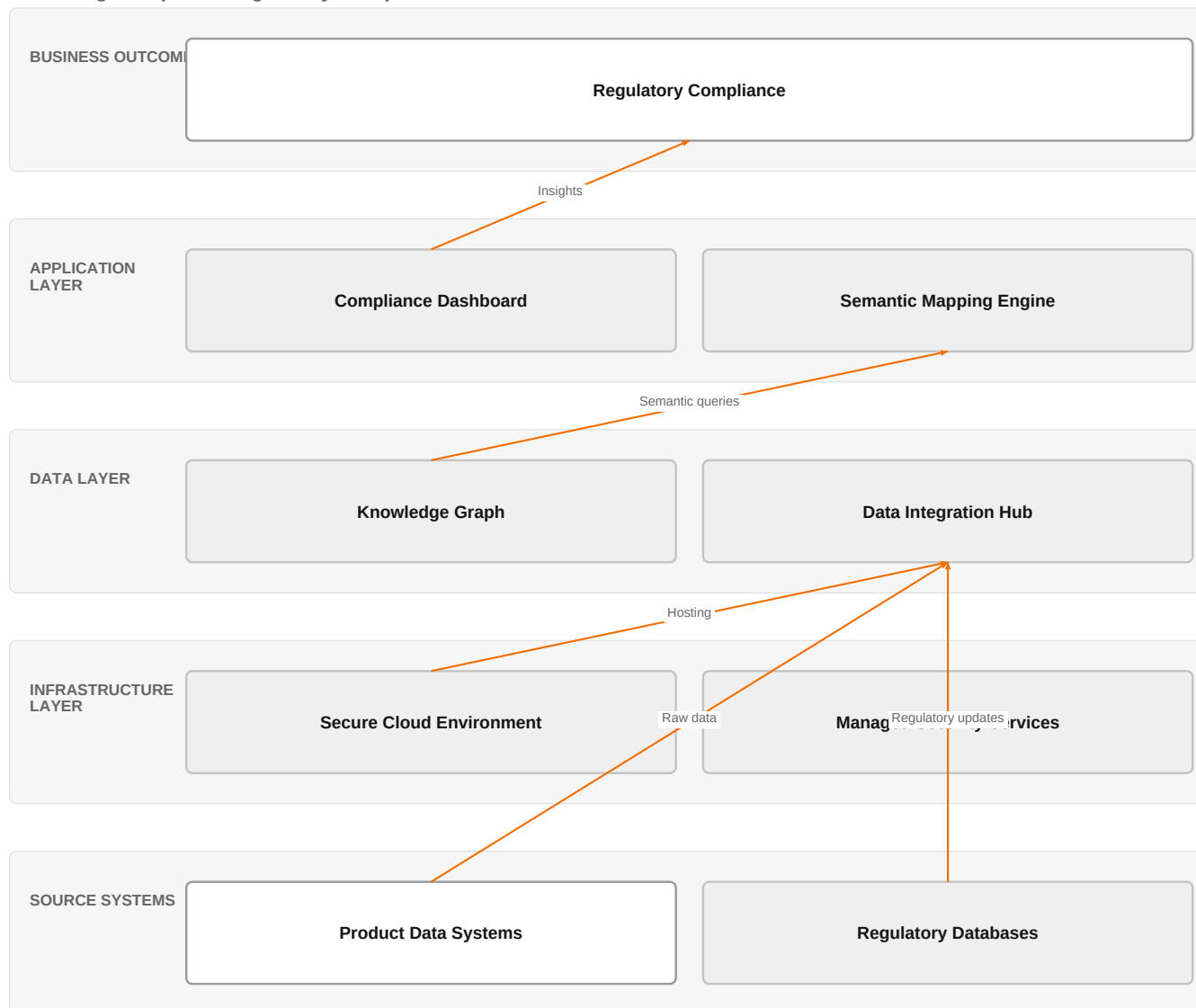
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€19.1m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19.2 026-07-08
Assumed obtainable share of the serviceable market	0.4 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 3 matching notices, matched on use case via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly

in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Knowledge Graph for Regulatory Compliance



Orange asset Partner Customer-owned Third party / to be sourced

This diagram shows how Orange's knowledge graph solution integrates product and regulatory data to deliver compliance insights.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU has adopted a regulation that aligns approval criteria for active substances in plant protection products with the new chemicals classification. This means companies must re-evaluate their existing dossiers and adapt their compliance processes. The regulation is not yet applicable, creating a preparation window. This regulatory shift demands a systematic approach to manage the semantic alignment between old and new criteria.

Sources: SIG-E250772CD266 ec.europa.eu 2026-06-30

Who buys, and why now

The chief regulatory officer or head of regulatory affairs at a crop protection or agrochemical company signs off. The pain is felt by regulatory affairs teams who must manually map thousands of data points across dossiers to new classification criteria. The trigger is the upcoming applicability date, forcing them to ensure every active substance approval remains valid.

Why it is hot now — every claim bound to a dated source

- EU regulations align approval criteria for active substances in plant protection products with new chemicals classification. [SIG-E250772CD266]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a team of AI, Data, and Cloud experts, along with Digital and Healthcare experts, to build a knowledge graph solution. The engagement starts with a discovery workshop to understand the customer's data landscape, then designs a semantic model that maps regulatory criteria to product data. Orange would deploy the solution on a secure cloud environment, leveraging Orange Cyberdefense managed services for protection. The solution would be validated with Orange Group researchers to ensure accuracy.

Why Orange can win

Orange brings a unique combination of healthcare domain expertise and advanced data capabilities. The HDS certification demonstrates our ability to handle sensitive health-related data, which is relevant given the health and environmental aspects of plant protection products. Our managed security services add trust, and our research arm ensures technical depth. However, our assets are not specifically tailored to agrochemical regulatory compliance, so we must position this as a data engineering and security play.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
HDS (Hebergeur de Donnees de Sante)	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
MicroPort CardioFlow	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

LOW competitive intensity (score 2.8; 0 of 8 listed players appear in this topic's own evidence)

One or two adjacent players. Orange would be shaping the space rather than fighting for it.

Competitor	Category	Position here	Basis	Evidence
IBM	Global systems integrator	sells Knowledge graph and semantic modelling; competes in Cloud	structural	—
Palantir	Data and AI platform	sells Knowledge graph and semantic modelling; competes in Cloud	structural	—
AWS	Hyperscaler	competes in Cloud — also an Orange partner	structural	—
Google Cloud	Hyperscaler	competes in Cloud — also an Orange partner	structural	—
Microsoft	Hyperscaler	competes in Cloud — also an Orange partner	structural	—
Accenture	Global systems integrator	competes in Cloud	structural	—
Atos / Eviden	Global systems integrator	competes in Cloud	structural	—
Capgemini	Global systems integrator	competes in Cloud	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How are you currently tracking the alignment of your active substance dossiers with the new EU chemical classification?
- 2 Which specific data sources and systems hold your approval criteria and product data?
- 3 Who in your organization is responsible for regulatory compliance reporting, and how do they currently manage changes in regulations?
- 4 Have you already started any projects to adapt to the new classification, and if so, what tools are you using?
- 5 What is your biggest pain point in ensuring compliance with the upcoming regulation: data mapping, semantic alignment, or reporting?
- 6 Are you looking for a one-time remediation or an ongoing compliance management solution?

Objections you will meet

They will say	You can answer
We already have a data team that can handle this internally.	That's a valid point. However, our strength is in combining healthcare domain expertise with advanced data engineering, and we can bring a proven methodology and security framework that might be hard to replicate internally. We can also augment your team rather than replace it.
IBM and Palantir have more mature knowledge graph platforms.	They do have strong platforms. But we offer a more integrated approach with managed security and healthcare compliance expertise, which is critical given the health-related aspects of plant protection products. We can also provide a more tailored solution that fits your existing cloud strategy.
This is just a regulatory change; we can handle it with spreadsheets.	For a small number of substances, spreadsheets might work. But with the scale of data involved and the need for ongoing alignment, a knowledge graph provides a more sustainable and auditable solution. It also enables better reporting to regulators.

Risks and unknowns

The main risk is that the customer may prefer a specialized knowledge graph platform from IBM or Palantir. We also lack a reference in agrochemical compliance, which could raise doubts. The exact scope of the regulatory alignment is not fully detailed in the evidence, so we may need to invest in regulatory expertise. Additionally, the timeline for applicability is unknown, which could affect urgency.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a life sciences client, outlining a knowledge graph solution that integrates chemical classification data with approval criteria, and include MicroPort CardioFlow as a reference to demonstrate our experience in regulated industries.
Sales	In a conversation with a regulatory affairs director at a European agrochemical company, mention Orange's expertise in data management and cybersecurity, referencing the HDS certification and Orange Cyberdefense managed services as proof of our ability to handle sensitive regulatory data.
Strategist / Innovator	Commission a deep-dive brief on the regulatory timeline and data-modeling requirements for aligning active substance approval criteria with new chemical classifications, leveraging Orange Group researchers and AI/Data experts to assess the feasibility of a knowledge graph solution.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-E250772CD266	2026-06-30	ec.europa.eu	1	Alignment of approval criteria for active substances in plant protection products with new chem...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (6 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:39+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.